UNIT 4 · TOPIC 12

Operation and Maintenance of Tractors and Power Tillers

Syllabus topic: Operation and maintenance of tractors and power tillers

Golden rules

1. Follow the operator's manual schedule (typical intervals below).
2. Service the **AIR CLEANER** — the single most critical item in dusty Indian fields. A choked cleaner → black smoke and power loss; dust ingestion → rapid wear ("dusting").
3. Handle fuel clean and sealed: 10-ppm BS-VI diesel + CRDI = zero tolerance for dirt and water.
4. Set correct tyre pressures and ballast for the job (traction economics, Unit 6).
5. Record hours, fuel and repairs (cost control, Unit 3, Topic 9).

Typical maintenance schedule (follow the operator manual)

Interval	Jobs
DAILY (10 h)	Fuel, engine oil level, radiator water, air cleaner (dusty: clean daily), tyre check, leaks
50 h	Grease all points, battery electrolyte, fan-belt tension, wheel nuts, oil-bath cleaner oil
250 h	Engine oil + filter change, fuel filter service, valve clearance check, tyre pressure / rotation
500 h	Injector check, transmission/hydraulic oil levels, steering & brake adjustment
1000 h	Transmission-hydraulic oil & filters, FIP calibration check, dynamo/alternator, full service

Power tillers

A **7–12 hp** single-axle walking tractor with the ROTARY TILLER as its main tool — dominant in wetland paddy and hill farming.

- **Maintenance** parallels the tractor: daily oil/air/water checks, V-belt tension, rotary-blade wear and bolts, transmission oil per schedule.
- **Operational cautions:** handlebar shake (hand-arm vibration exposure — Unit 5); cage wheels for puddling; decompression lever for hand-crank starting.

Off-season storage

1. Clean and wash the machine.
2. Drain or fill the fuel, depending on storage duration.
3. Run the engine to distribute preservative oil.
4. Slacken the belts.
5. Block the tyres off the ground.
6. Grease exposed metal.
7. Remove the battery and charge it periodically.
8. Cover the machine in a dry shed.

This prevents the corrosion and sulphation losses that cause season-start failures.

Practice MCQs

Q203. In dusty Indian conditions, the single most critical daily maintenance item is:

(a) Seat cushion
(b) Air cleaner service
(c) Horn
(d) Mirror

Q204. Engine oil and filter are typically changed every:

(a) 10 h
(b) 50 h
(c) 250 h
(d) 2000 h

Q205. Transmission-hydraulic oil and filters are typically changed at:

(a) 100 h
(b) 250 h
(c) 500 h
(d) 1000 h

Q206. Fan-belt tension and greasing of points are done about every:

(a) 10 h
(b) 50 h
(c) 500 h
(d) Never

Q207. Power tillers commonly fall in the power range:

(a) 2–4 hp
(b) 7–12 hp
(c) 20–30 hp
(d) 40–50 hp

Q208. The principal working attachment of a power tiller is the:

(a) MB plough
(b) Rotary tiller
(c) Baler
(d) Sprayer

Q209. For long off-season storage, the battery should be:

(a) Left connected
(b) Removed and charged periodically
(c) Drained of acid and left dry forever
(d) Frozen

Q210. Dust ingestion past a faulty air cleaner causes:

(a) Better cooling
(b) Rapid abrasive wear of liners and rings ('dusting')
(c) Higher octane
(d) Cleaner oil

Answers

Q203 — (b) Air cleaner service
Q204 — (c) 250 h
Q205 — (d) 1000 h
Q206 — (b) 50 h
Q207 — (b) 7–12 hp
Q208 — (b) Rotary tiller
Q209 — (b) Removed and charged periodically
Q210 — (b) Rapid abrasive wear of liners and rings ('dusting')

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units (2nd ed., 1989), Ch. 14 "Tractor Tests and Performance", pp. 403–446; MoRTH TREM notifications (as verified in the source notes, July 2026); CFMTTI Budni documentation; Jagdishwar Sahay (maintenance schedules); NPTEL lectures.